sometimes come and visit us, as Halley's Comet did most recently in 1910 and 1986. Comets are made mainly of ice, plus smaller amounts of rocky and organic material. When heated, the ice vaporizes, forming the long and lovely tails blown outward by the solar wind and the pressure of sunlight. After many passages by the Sun, the ice is all evaporated, sometimes leaving a dead rocky and organic world. Sometimes the remaining particles, the ice that held them together now gone, spread out in the comet's orbit, generating a debris trail around the Sun.

Every time a bit of cometary fluff the size of a grain of sand enters the Earth's atmosphere at high speed, it burns up, producing a momentary trail of light that Earthbound observers call a sporadic meteor or "shooting star." Some disintegrating comets have orbits that cross the Earth's. So every year, the Earth, on its steady circumnavigation of the Sun, also plunges through belts of orbiting cometary debris. We may then witness a meteor shower, or even a meteor storm—the skies ablaze with the body parts of a comet. For example, the Perseid meteors, seen on or about August 12 of each year, originate in a dying comet called Swift-Tuttle. But the beauty of a meteor shower should not deceive us: There is a continuum that connects these shimmering visitors to our night skies with the destruction of worlds.

A few asteroids now and then give off little puffs of gas or even form a temporary tail, suggesting that they are in transition between cometdom and asteroidhood. Some small moons going around the planets are probably captured asteroids or comets; the moons of Mars and the outer satellites of Jupiter may be in this category.

Gravity smooths down everything that sticks out too far. But only in large bodies is the gravity enough to make mountains and other projections collapse of their own weight, rounding the world. And, indeed, when we observe their shapes, almost always we find that small worldlets are lumpy, irregular, potato-shaped.

THERE ARE ASTRONOMERS whose idea of a good time is to stay up till dawn on a cold, moonless night taking pictures of the sky—the same sky